

SMART INDIA HACKATHON 2025

- **Problem Statement ID – SIH25029**
- **Problem Statement Title - Authenticity validator for Academia**
- **Theme - Smart Education**
- **PS Category- Software**
- **Team ID -**
- **Team Name - Lumora**



PROPOSED SOLUTION

Student

- ➔ Upload Document
- ➔ Receive a unique verification token as proof.

Recruiter

- ➔ DigiLocker (API verification).
- ➔ Token verification (already verified)
- ➔ Doc upload & verification pipeline

Verification pipeline

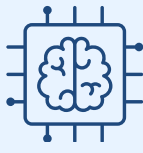
1. OCR + Database matching.



2. Blockchain-based validation



3. AI/ML forgery detection

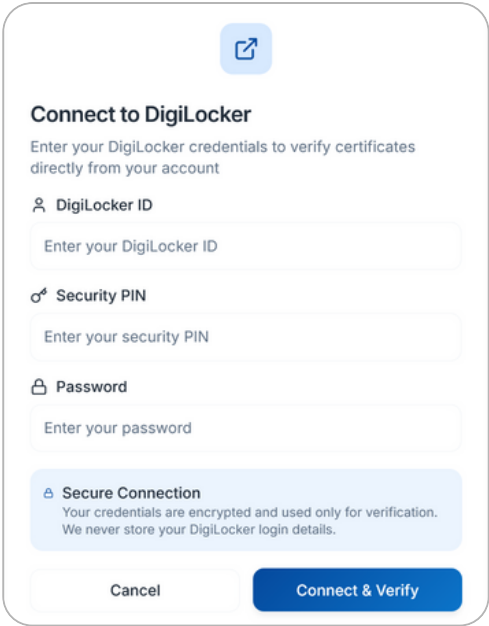
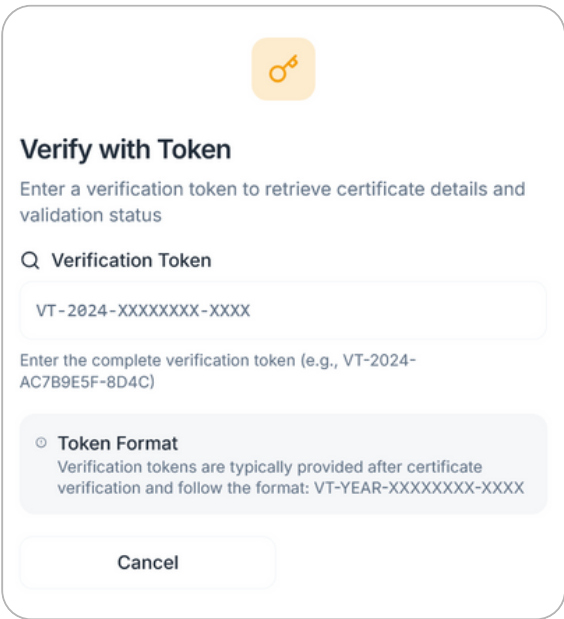
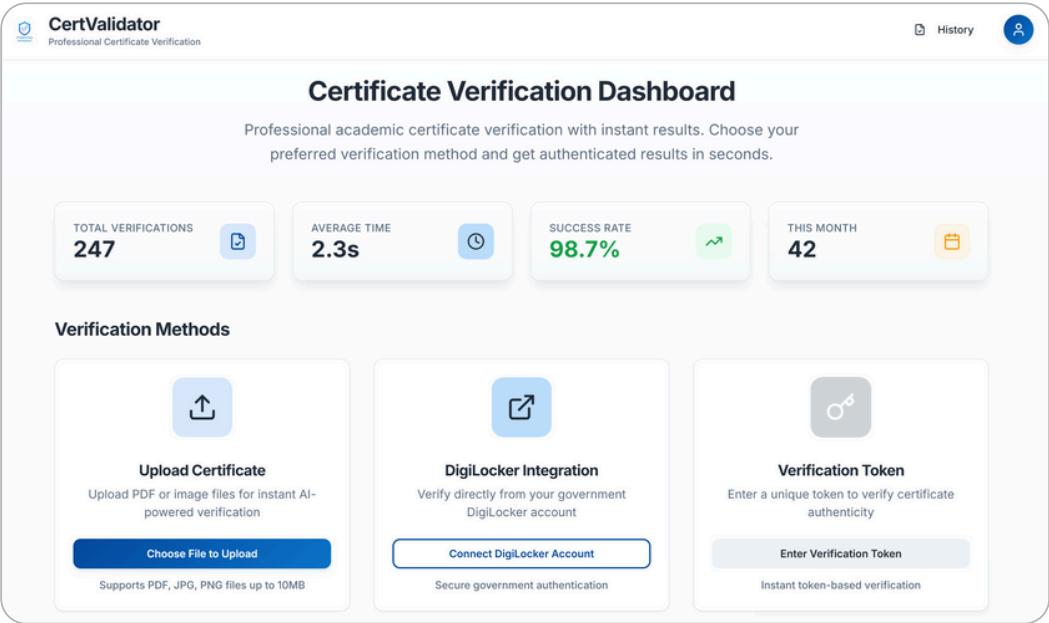


ADRESSING PROBLEM

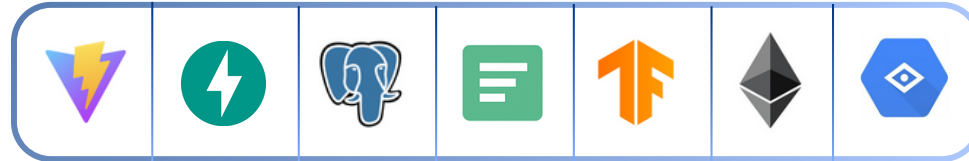
- Stops fake/forged certificates with **multi-step checks**.
- Gives recruiters quick and reliable verification.
- **Cuts delays** of manual inspections.
- Ensures **uniform verification** across institutions.
- **Recruiter verified** via GST/CIN, domain,cross-platform
- **Safeguards** academic integrity.

INNOVATION & UNIQUENESS

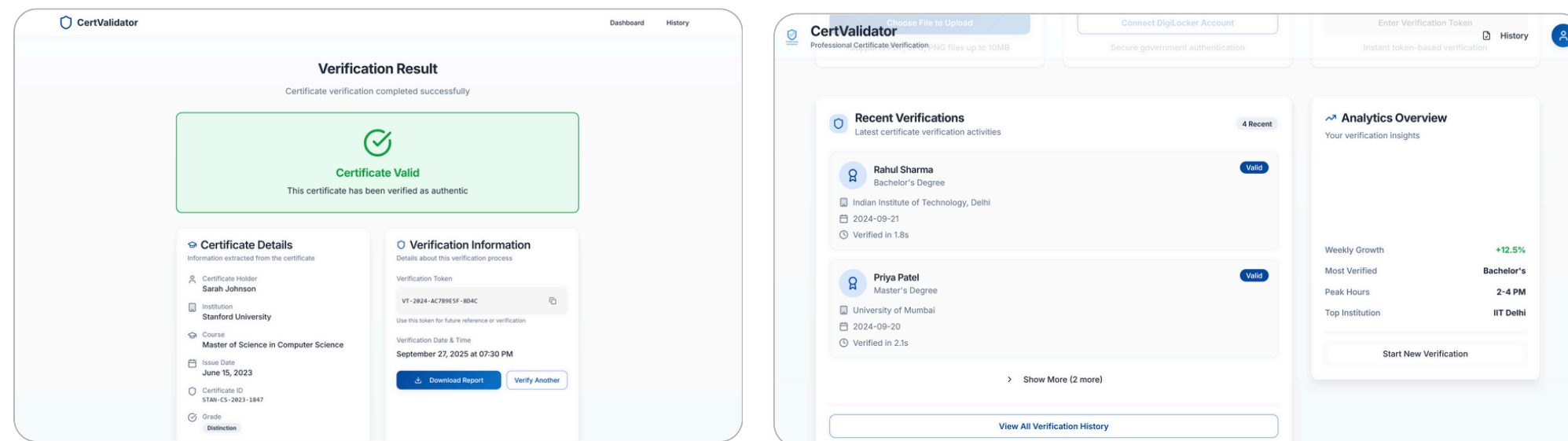
- Combines **3 Step** Verification for all **edge cases**.
- **Fallback paths** : manual review if OCR/DB fails.
- Token system for **reusable verification**.
- **Supports both old and new certificates**.
- **Scalable** for state/nationwide rollout.



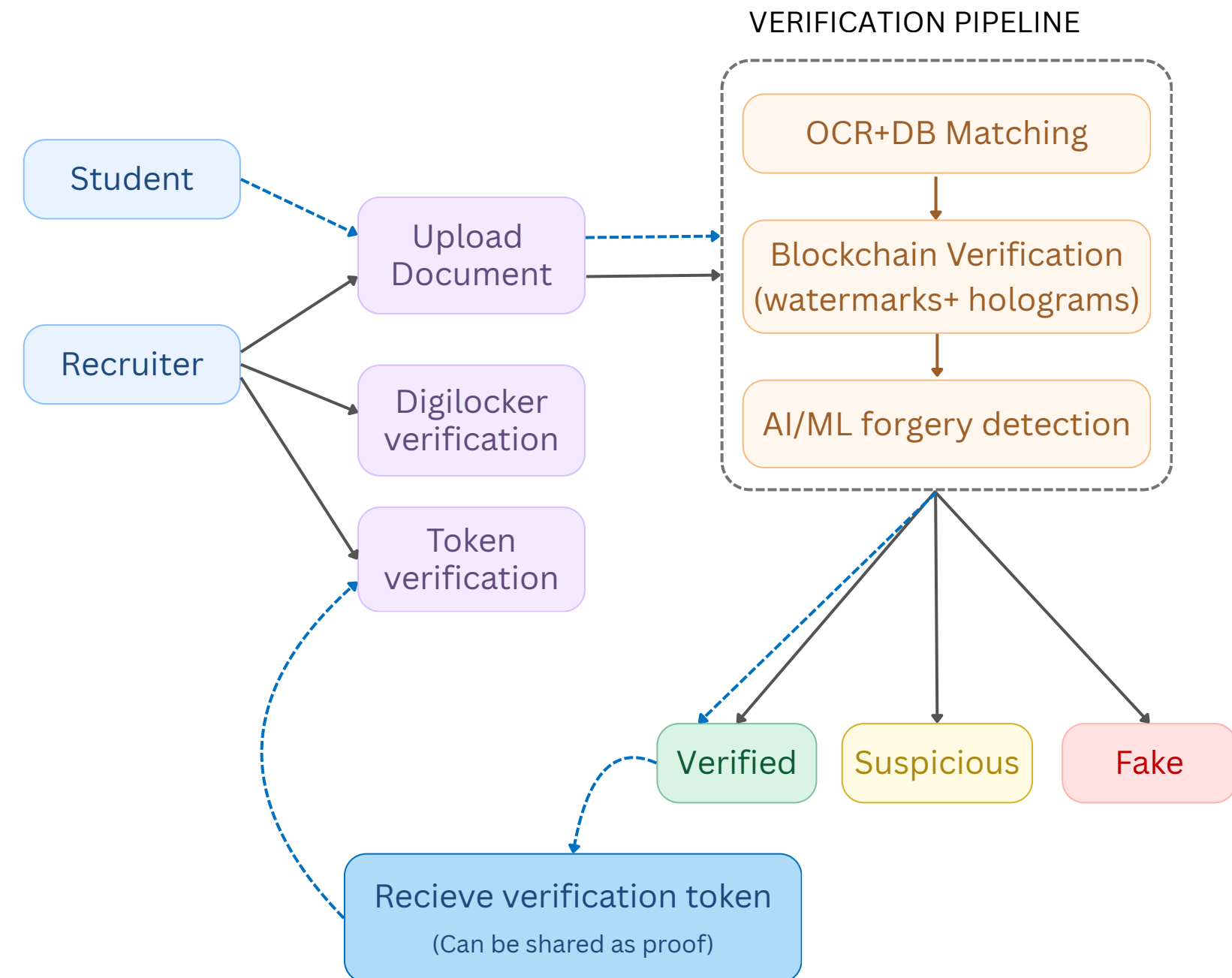
TECHNOLOGIES



Frontend	React.js
Backend	Python (Django / FastAPI)
OCR	Tesseract OCR / Google Vision API
Blockchain	Hyperledger / Ethereum smart contracts
AI/ML Models	CNNs+Transformers for forgery & anomaly detection
Database	MySQL / PostgreSQL for structured data
Integration	DigiLocker API for government-verified credentials



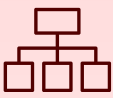
FLOWCHART



FEASIBILITY

-  **Technical:** OCR, AI/ML, and Blockchain are proven tools for secure verification
-  **Operational:** APIs and dashboards enable easy rollout for both digital & old records.
-  **Security:** Encrypted, access-controlled, and compliant with IT laws & GDPR.
-  **Scalability:** Cloud + private blockchain keeps costs low and supports nationwide use

CHALLENGES & RISKS

-  **OCR Errors :** Low-quality scans may cause text misreading.
-  **Legacy Systems :** Many universities still use old/manual records.
-  **Blockchain Cost :** Public blockchain can be slow and expensive.
-  **Data Privacy:** Student data is sensitive and must be well-protected.

MITIGATION STRATEGIES

-  **Fallback :** Manual review for poor scans or OCR errors.
-  **Cost Control :** Use private blockchain (e.g., Hyperledger) to cut costs.
-  **Security :** AES-256 encryption + access control + regular audits
-  **Adoption :** Start with pilot universities, then expand gradually.

IMPACT



Institutions (Colleges/Universities) → Protect academic reputation, reduce fake certificate misuse.



Recruiters & Employers → Faster and more reliable candidate verification.



Students → Genuine achievements recognized fairly, no competition with fake degree holders.



Government & Agencies → Easier fraud tracking, supports Digital India & education reforms.

BENEFITS



Trust & Transparency → Builds confidence in educational and hiring ecosystem.



Time & Cost Savings → Eliminates delays of manual checks and reduces administrative costs.



Fraud Prevention → Detects and blocks forged/tampered certificates early.



Fairness & Integrity → Creates equal opportunities for deserving candidates.

CASE STUDIES

- Large-Scale Fraud in Uttar Pradesh: In July 2025, the Supreme Court flagged large-scale fake certificate fraud in Uttar Pradesh, with reports indicating that 8,000 certificates from Agra University were found to be fake. *Source : TOI*
- 17 teachers in Jharkhand were dismissed for using invalid certificates to secure their positions. The authorities took strict action against these individuals to maintain the integrity of the education system *Source : TOI*

RESEARCH PAPER

- Shivam Gangwar, Anushka chaurasia, “Blockchain-based Authentication and Verification System for Academic Certificate using QR Code and Decentralized Applications.” International Journal of Computer Applications (0975- 8887) Volume 186- No.26, June 2024.
- Rane, Mukul & Singh, Shubham & Singh, Rohan & Amarsinh, Vidhate. (2020). Integrity and Authenticity of Academic Documents Using Blockchain Approach. ITM Web of Conferences. 32. 03038. 10.1051/itmconf/20203203038.
- Singh, Rohan. “Integrity and Authenticity of Academic Documents Using Blockchain Approach.” ITM Web of Conferences, EDP Sciences.

Nearly 28% of job applicants submitted fake degrees, highlighting widespread academic fraud (2020 report).